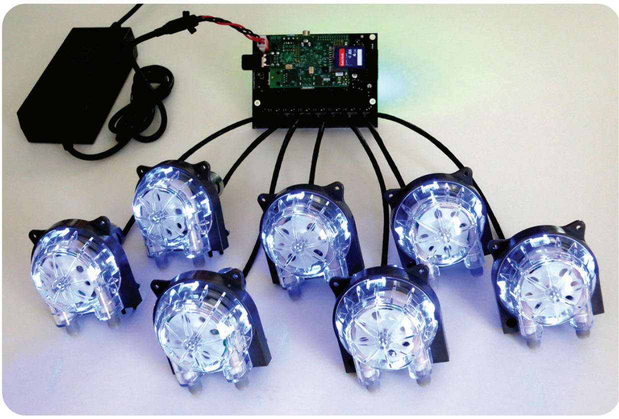
Bartendro™ B7 / B15 Kits

Getting Started Guide

Model Nos.
B107V1
B115V1



KICKSTARTER EDITION

Thanks for backing us!

Please Read Before Operating

Congratulations!

You are one of the first owners of a new enabling piece of technology. These precision Bartendro™ Dispensers and electronics are the same as what go into Bartendro™ cocktail bots. The applications don't stop at cocktails though, people have been coming up with many applications from hydroponics to soap making. What will you come up with? Share your projects at:

www.partyrobotics.com/community/sharing

⚠ Important Precautions

- Not intended for medical use.
- Electronics may be sensetive to static dissipation. Observe proper precautions for handling.
- It is not recommended to keep clear tubing immersed in liquid for extended periods of time.
- Beige tube (inside the Dispensers) life is approximated to be 500 hours of use.
- If storing long term without use, remove beige tubing from dispensers to extend tube life.
- Dispensers may need to be periodically re-calibrated as tubing wears.
- Please drink and use responsibly.

Specifications

Max Flow rate: 700mL / min
Motor Operating Voltage: 24V
Minimum Dispense Volume: 2mL
Repeatability: ±1mL
Dispenser Current Consumption at 24V: 350mA Typical, 500mA max
System Current Consumption at 24V: 0.5A Idle, 3A All dispensers on (B7)
0.5A Idle, 6A All dispensers on (B15)
Max Cable Length to Router Board: 10ft

What's Included:

B7	B15	Part No.	Item
7	15	A007V1	Bartendro™ Dispenser With Liquid Sense (includes two compression nuts)
1	1	M101V1	A Chopstick (tubing replacement tool)
1	1	A017V1	Bartendro™ Router Board (with hardware)
1	1	E004V1	Raspberry Pi (RPI)
1	1	E007V1	SD Card
1	1	E005V1	USB WiFi Dongle
1	1	E009V1	24V, 200W Power Supply with Power Cord
1	1	A014V1	Power Adapter Cable
1	1	TXXXV1	Food Grade Tubing (about 5ft per dispenser)
4	8	M102V1	Standard Bottle Topper
2	5	M103V1	Large Bottle Topper
1	2	M104V1	Extra Large Bottle Topper
7	15	M105V1	Bottle Topper Cap
7	15	M033V1	Tubing Keeper
7	15	A009V1	Liquid Sensor Adapter

Quick Start Instructions

You will need:

AC Outlet
Device with Web Browser (phone, tablet, laptop, etc)
Wired Networking Router (for software updates)
Computer (for software updates)

Hardware Setup:

1. Make sure the Raspberry Pi is securely connected to the Router Board and is supported with standoffs. Take care not to damage the header if removing the board. (gentle rocking works best)

2. Attach the provided SD Card and USB WiFi dongle to the RPI.

3. Connect Dispensers to the Router Board using the provided ethernet cables. Note that ethernet cables are only used for convenience and commonality, the ethernet protocol is not used.

4. Port numbers on the Router Board should be noted when plugging into a new Dispenser. The numbers can optionally be written down in the white box on the back of each dispenser.

5. Connect the Power Adapter Cable to the Router Board and the Power Supply to an AC outlet. Mate both ends of the 4 pin connectors, the Raspberry Pi will begin to boot.

6. Once the Dispenser's LEDs begin to fade in and out (about 1 minute), the system is ready for use. If they don't get to this state, see the Troubleshooting section on page 8.

7. Mount the dispensers appropriatly. See dimensions on page 7.

8. Attach clear tubing to the dispensers using the provided compression nuts. Make sure to budget about 5ft per pump. Cut an angle into the bottom of the Dispenser's input tubing to prevent it from attaching to the bottom of the bottles. See video at partyrobotics.com/docs.

9. Clear tubing cannot be attached directly to liquid sensors, so use Liquid Sensor Adapters in between making sure to use the port that is farthest from the board. Use the black Tubing Keepers to keep both the Dispenser input tubing and liquid sensor tubing neatly together.

Getting Started with the User Interface

Connect to your bot:

1. Using a WiFi enabled device, find the “bartendro” WiFi network by looking for it in your settings and connect to it.

2. Once connected (up to ~30 sec the first time) you will need to go to a web browser, such as Chrome, and type in any web addess. You will automatically be redirected to the Bartendro user interface (UI).

3. The main menu may or may not have drinks populated. To tell the bot which ingredients are present, you will need to log into the admin screen. You may do this by either searching for the hidden link at the bottom left corner of the main menu, or typing bartendro/admin in the address bar of your browser.

When prompted use the default:

Name: bartendro
Password: boozemeup

Note: Your login name and password can be modified by changing a file on the Raspberry Pi. Instructions on page 7.

Once connected to Bartendro you will have access to the following:

Admin

Admin - Dispensers: Controls ingredient selection for each dispenser from a drop down menu. Ingredients come from the “booze” tab. Choose wisely, your drink options depend on it! Note that taking manual control comes with responsibility. The “clean” button will indescriminitely dispense all of your booze if pressed by accident and you will be a sad panda. Each dispenser can indivudially be toggled on and off by using the “turn on”/”turn off” buttons and small increments can be dispensed using the “dispense xxmL” button (helpful when priming). The “liquid levels” button forces a check of liquid levels, but this only works if the feature is enabled. See page 7.

Admin - Drinks: Displays the drink list and allows you to make your own custom custom recipes. Select “add booze” and pick from the list of ingredients. The minimum number of parts is 1 and parts should be whole numbers, or will otherwise be rounded. All fields are required. Make sure to press save for your drink to be stored.

Admin - Booze: Displays the ingredient options and allows you to add your own custom ingredients. Adding the type, whether alcohol or sweetness / tartness will allow for customizations in the drink menu.

Admin - Liquid Out: Liquid sensing is disabled by default. To turn this feature on, follow the steps on page 7. Once enabled, each dispenser can be tested by inserting the liquid sensor tubing into a shot glass of water. Type “90” as the threshold and press “go”. Slowly lift the sensor tubing out of the water to see where the “out” level is. This threshold can be changed in software.

Main Menu

Menu: This is where your drinks are available for selection. The list automatically updates depending on what is loaded in the dispensers screen. Note that the menu may appear differently depending on the number of dispensers detected.

Menu - Drink Customization: After you select your drink from the menu a second screen will display allowing you to customize your drink. You will have the option to modify size, and depending on the ingredients, strength and sweeterness / tartness.

Cleaning and Maintenance

All tubing should be cleaned regularly.

1. Empty what remains in the tubes by turning on the dispensers. You may use the “turn on,” “dispense xxmL” or “clean” button to do this.

2. Fill a pitcher with hot soapy water and place the input tubes in it after they’ve been drained. Press the “clean” button on the admin > dispensers screen and and this will cycle through each dispenser.

3. Repeat step 2 but with clean water in the pitcher. Sanitizing solutions may also be used.

4. Finally, run the clean cycle again with no liquids at the input. This will empty the tubes of liquid and prepare it for its next use.

The inner, beige, dispenser tubing should be replaced after 500 hours of use, and removed if storing for prolonged periods of time.

Getting under the hood:

There are several settings that aren't exposed through the UI yet but can be changed by modifying the config.py file on the Raspberry Pi. Things like: use of liquid level sensors, display units, default size, taster buttons, the requirement to be logged in to dispense, etc.

To modify these settings:

1. Plug an ethernet cable from the RPI into a wired networking router. We've dissabled the ability to do this wirelessly for security reasons.

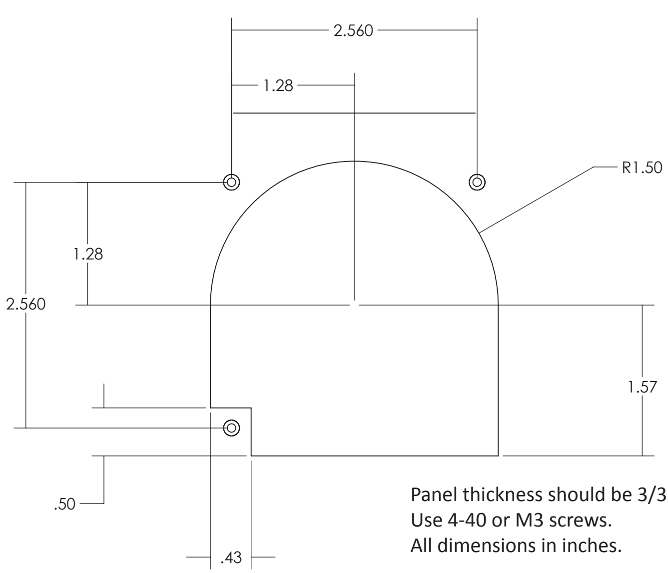
2. Use a terminal window on a Mac or Linux box, or use a program like putty.exe in Windows to SSH into the bot. You may want to log into your networking router to discover the RPI's IP address.

Once connected,
login as: bartendro
password: hackme!

You can use vi to edit:
bartendro@bartendro ~ $ vi bartendro/ui/config.py

Save the file and power cycle for changes to take effect.
See partyrobotics.com/docs for more detail.

Dispenser Mounting Interface



Panel thickness should be 3/32" or less.
Use 4-40 or M3 screws.
All dimensions in inches.

Troubleshooting

Dispenser LEDs are blue and /or green after power up

This is normal, LEDs will remain in this mode until proper communication with dispensers is established (about 1 min). If not getting there, make sure SD card is properly seated

Dispenser LEDs are flashing red

A current limit has been triggered on a dispenser, press the “reset” button in the admin > dispensers page to restore proper operation. Check tubes for blockage.

Air bubbles in tubes, or dripping

This is a sealing issue. Make sure tubing properly protrudes from the grey compression nuts by about 1/4”. The chopstick tool is handy for this. Make sure the compression nuts are tightened so that only one or two threads are visible.

Drinks not showing up in Main Menu

Check to make sure an ingredient is selected and saved for each dispenser and that there is a drink that uses those ingredients. If liquid level sensors are enabled, and the liquid level tubes are not inserted in liquid, or are in too little of liquid, the menu will be blank.

Still Having Issues?

Try searching the forums at partyrobotics.com/forums. Post your questions so that others may learn from your issues. If stuck, email us at: support@partyrobotics.com

More documentation at:

www.partyrobotics.com/docs

Warranty

We’re here for you! If you have made modifications to your hardware or software and are having technical issues, we will be glad to attempt to resolve them with you. If an item was damaged in shipping, or is otherwise defective or malfunctioning, please email us right away and we will do our best to make things right on a case-by-case basis.